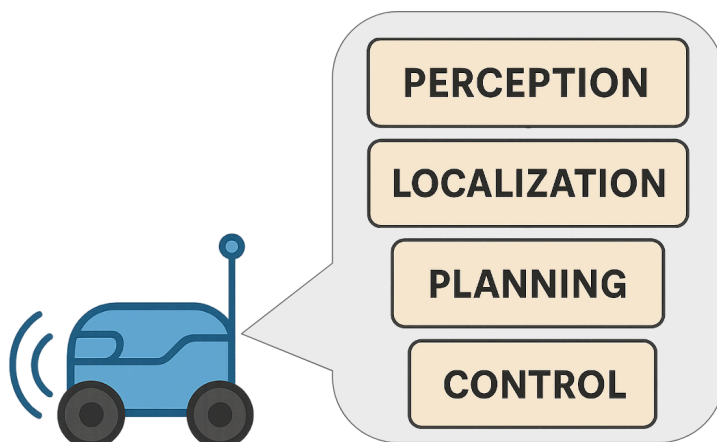


Syllabus: Introduction to Robot Autonomy



Instructor:

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Course Description:

This course will explore the major elements of the autonomous stack of a robotic system: perception, localization, planning, and control. The course will also introduce students to the Robot Operating System (ROS), robot simulation environments, and the basics of utilizing robot hardware. Throughout the duration of the semester, students will construct a basic autonomy stack in ROS, evaluate their implementation in a simulated environment, and deploy their autonomy stack on physical hardware as a proof of concept.

Course Learning Outcomes (CLOs):

By the end of this course:

- Students will create their own ROS packages that make up the main elements of a robot autonomy stack.
- Students will assemble a vision module that can find transformations between two laser scans for a robot with a LIDAR using the Iterative Closest Point algorithm.
- Students will develop a particle filter for localizing a robot in a known environment given a previous robot state, odometry measurements, and a transposition from visual data.
- Students will develop a path planning algorithm of their choice for holonomic robots that finds a collision-free path from a start location to a goal location.

- Students will design a PID controller for a holonomic robot to develop an ego-planner that follows a predetermined trajectory.
- Students will construct a complete autonomy stack by combining their lab assignments on Perception, Localization, Control, and Planning and demonstrating how these parts function together to control a robot.

Inclusive Teaching Statement:

I am committed to creating an inclusive learning environment that values diversity and fosters a sense of belonging for all students. Diverse classrooms enrich the learning experience of all students and I will strive to ensure all voices are heard and respected. There will be three anonymous surveys conducted throughout the course: one during the first week to gauge student backgrounds, one during the fourth week to get early student feedback on the course, and a one during the last week to get student feedback at the end of the course.

All required course material will be provided to the students and accessible online. This course requires access to a computer. Loaner laptops are available from the Computer Science department and students who do not have a personal computer are strongly encouraged to get in contact with me so that we can arrange suitable accommodations for the student to continue in the course without interruption.

Student participation is welcomed and strongly welcomed in this course. I am committed to being available to all students and students are encouraged to reach out for any matter pertaining to the course or professional and personal development. I have an open door policy for all matters (both within and outside of the course) and students are strongly recommended to contact me if they feel that there exists a barrier preventing them from participating in the course to their full potential at any point during the semester.

Course Policies:

In addition to policies and regulations on academic integrity from the university, this course only has one policy: Students are expected to ensure that their actions during the class do not take away from the learning experience of other students. If at any point a student feels that one of their peers is violating this policy, they are strongly encouraged to discuss the violation with the course instructor.

How to Succeed in this Course:

To be successful in this course, students will need to have an open mindset to learning, anticipate challenges, and diligently seek out help when needed. I will be available during my office hours and during class lab sessions to support students and provide feedback. The course is centered around multiple projects, done in both group and individual settings, that are supported by lectures. Start early and start often. Attendance is not strictly required, but is strongly encouraged. The projects are designed to reinforce student understanding of the course materials while giving students experience building and debugging a basic robot autonomy stack. By their

nature, the projects will be challenging at times and students are welcome to ask questions and seek out help when stuck.

Students will be working in a group setting for course projects in this course. Students are strongly encouraged to capitalize on this experience and put forth the same effort that they expect from their fellow group members. Being an active member and a good groupmate are essential to being successful in this course. If at any point a student feels that one or more of their group members is failing to meet this expectation or a student feels excluded from their group, they are strongly encouraged to speak with the instructor.

Assessment Plan:

Students will be assessed in the following ways.

- Lecture Worksheets – Students will complete a written worksheet that accompanies introductory lectures. The worksheets will serve as a formative assessment to gauge student understanding, encourage engagement with the lecture material, and to reinforce concepts introduced during lectures.
- 60-Second Essays – After lectures on advanced topics, students will write a 60-second essay describing, at a high level, the material covered in class.
- Quiz – Students will take a quiz on new introductory-level material to gauge their understanding of the topics and to reinforce the material covered in lecture.
- Labs:
 - Lab 0 (ROS) – Students will develop a basic ROS package that includes: publisher nodes, subscriber nodes, services, and a launch file.
 - Lab 1 (Perception) – Students will develop a ROS package that parses LIDAR scans and finds transformations from one frame of reference to another using the Iterative Closest Point algorithm.
 - Lab 2 (Localization) – Students will develop a ROS package to localize a robot within a known map using a filtering method of their choice.
 - Lab 3 (Planning) – Students will develop a ROS package that finds a trajectory for a robot to follow to navigate between two points in a known environment while avoiding obstacles using a path planning algorithm of their choice.
 - Lab 4 (Control) – Students will develop a ROS package that runs a simple ego-planner using a PID controller and a given trajectory.
- Term Project – Students will integrate all labs into a single framework that performs the four main elements of a robot autonomy stack on a provided robot platform. Students may incorporate feedback from their lab submissions and integrate them into the project Milestones for extra credit toward points lost on the labs. This assignment will be completed in five steps over the course of the semester:
 - Initial Setup (Milestone 0) - Teams will integrate their basic ROS package from Lab 0 on the robot and demonstrate that they can remotely control the robot using a controller.
 - Perception (Milestone 1) - Teams will integrate their ROS package for perception from Lab 1 on the robot platform.

- Localization (Milestone 2) - Teams will integrate their localization ROS package from Lab 2 on the robot platform.
- Planning (Milestone 3) - Teams will integrate their path planning ROS package from Lab 3 on the robot platform.
- Control (Milestone 4) - Teams will integrate their ego-planner ROS package for perception from Lab 1 on the robot platform.

Notes/Reflections:

I think it would be helpful for me to create a list of individual topics that I want to cover in lectures, then create a basic schedule and see how well I think I can fit all of my assessments into the plan. I think this would help me decide if something is missing from the course.

I found writing the “Inclusive Teaching Statement” pretty hard. I’m still not sure that what I wrote is that great so I’m eager to get more feedback on this. I did find the UMass page in the Course Syllabus Resources Library helpful. For course policies, I have usually seen this section filled with boiler plate text from the university on academic integrity and Title IX type language that every professor puts into their syllabus. I’m guessing that this stuff is sometimes required by the university and acts as a “covering my ass” for the instructor, but I have usually found that stuff to just fill a syllabus with extra content that no one really reads and takes away from the other parts. Maybe it would be good to include a separate document with the syllabus that covers all of this stuff and reference that separate doc from in the syllabus. This would keep the syllabus clean while also providing that extra stuff. Maybe an appendix..?

I am pretty happy with my CLOs and assessments. I know that the written assessment part is a little vague, but I intend for this to be something that goes along with the major CLOs that checks understanding at a high level. I felt that it would be excessive to list out the same a bulleted list of written assessments similar to the labs because this would just be a repeat of the labs and these assessments are not meant to be as important as the labs (which is why I listed out the labs but left written assessments as a single entry).